SMART HEALTH RISK INDICATOR (SHRI)

AI-Powered Multi-Domain Clinical Decision Support System | Major Project Review Document

Project Phase: 75% Review Completed & 25% Final Defense Roadmap	**Team Size:** 3 Engineering Members
Core Stack: Python, Streamlit, Scikit-Learn, Plotly, NLP	**Local Web Portal:** http://localhost:8501

1. 75% Review Milestone: 3-Member Team Contribution Breakdown

For the 75% project review, the engineering workload was split into three modular, non-overlapping domains ensuring each team member has substantial technical ownership.

Team Member	Role & Core Deliverable	Key Technical Modules
■ **Member 1**	**ML & Ensemble Lead** Core Machine Learning & Uncertainty Engine	• 6,000 Hybrid Clinical Dataset Generator • 20D Multimodal Feature Matrix • DepNet, AnxNet, SleepNet (Specialists) • FusionNet Meta-Learner (23 Inputs) • Monte Carlo Dropout (95% CI, 20 passes)
■ **Member 2**	**NLP & XAI Lead** Clinical NLP, Crisis Alerting & Explainability	• Real-Time Clinical NLP Sentiment Engine • 90-Word Clinical Lexicon & Bigrams • Negation & Intensifier Multipliers • Permutation SHAP Feature Attribution • Immediate Crisis UI Escalation Modal
■ **Member 3**	**Full-Stack Lead** Web Platform, Forecasting & Interoperability	• Interactive Streamlit Web Application • Longitudinal Trajectory (LR + EWMA) • Statistical Anomaly Detector (2σ) • Evidence-Graded CBT Planner • JSON / FHIR Diagnostic Export

2. Individual Viva Speech Scripts for 75% Review

■ Member 1 (ML & Ensemble Architecture Lead):

"Respected Examiners, my primary contribution is the core Machine Learning Pipeline, 20-dimensional Feature Engineering, and the 4-Model Stacked Ensemble with Uncertainty Quantification. Rather than a single monolithic classifier, I designed 3 specialized sub-networks: DepNet (PHQ-9), AnxNet (GAD-7), and SleepNet (ISI/lifestyle), fused by FusionNet—a 23-input meta-learner. To eliminate overconfidence, I implemented Monte Carlo Dropout with 20 stochastic test-time passes, yielding a verified 95% Confidence Interval."

■ **Member 2 (Clinical NLP & Explainable AI Lead):**

"Respected Examiners, I developed the Real-Time Clinical NLP Sentiment Engine, Crisis Keyword Escalation, and Explainable AI (XAI) using SHAP. Adolescent self-reports contain unstructured emotional signals; I built a sentiment parser with a 90-word clinical dictionary, negation handling, and 20 high-risk bigrams that immediately trigger crisis hotlines like Tele-MANAS. Furthermore, I built a permutation SHAP module that explains to doctors exactly which features drove the score up or down relative to baseline."

■ **Member 3 (Full-Stack Platform, Forecasting & CBT Lead):**

"Respected Examiners, I engineered the interactive Web Application, Longitudinal Forecasting, Anomaly Detection, and CBT Interventions. I built the Streamlit clinician portal with real-time Plotly charts and gauge indicators. To track recovery, I created a 30-day trajectory forecast combining Linear Regression with Exponentially Weighted Moving Average (EWMA, alpha=0.4) to flag statistical spikes ($|z| >= 2\sigma$), and added automated CBT recommendations with FHIR/JSON export."

3. Five-Step Live Demonstration Protocol for Review

- **Step 1: Patient Assessment:** Load 'Pooja Patel (High Risk)' preset in the web portal. Show the composite score (74/100, Tier 4) and the 95% Confidence Interval badge.
- **Step 2: 4-Model & MC Dropout:** Open Ensemble tab. Show the Plotly Gauge Chart and the 20-pass Monte Carlo density histogram with shaded 95% CI region.
- **Step 3: SHAP Explainability:** Open SHAP tab. Show the horizontal waterfall chart decomposing score into positive risk factors (+18.2) vs mitigating habits (-6.1).
- **Step 4: Longitudinal & Anomaly:** Open Trajectory tab. Show the 30-day projected forecast line and the 2σ statistical anomaly alert box.
- **Step 5: Crisis NLP Escalation:** Type 'I can't cope anymore and want to end it all' in journal text. Show the instant red crisis escalation alert with emergency helplines.

4. The Remaining 25% Roadmap: Taking Project to 100%

To achieve 100% final defense readiness, the remaining 25% is partitioned into four concrete engineering deliverables with clear technical specifications:

Module	Scope & Weight	Technical Deliverable & Method
1. Clinical Validation	**10% Weight** Real-world benchmarking	• Generate multi-class ROC-AUC curves for all 5 tiers. • Compute Sensitivity (Target >92%) & Specificity (>88%). • 10-Fold Stratified Cross-Validation across age cohorts. • Brier Score & Platt Calibration curve calculation.
2. Audio Biomarkers	**5% Weight** Acoustic speech prosody	• Audio check-in ingestion widget via Python librosa. • Pitch jitter, shimmer & vocal micro-tremor extraction. • Speech cadence, pause latency & 13 MFCC coefficients. • Fuse audio score as 5th input branch into FusionNet.
3. Production API	**5% Weight** FastAPI & Docker	• Asynchronous FastAPI REST microservice (/predict, /shap). • Dockerfile & docker-compose containerization. • Interactive OpenAPI / Swagger documentation (/docs). • HL7 FHIR DiagnosticReport JSON interchange.
4. Regional Localization	**5% Weight** Multi-language & Offline	• Translate questionnaires into Hindi, Telugu, and Tamil. • Multi-lingual sentiment lexicon & transliteration. • Offline Progressive Web App (PWA) sync for rural PHCs.

5. Examiner Defense Script for the Pending 25%

Question from Examiner: ***"Your 75% is working. What is pending in the remaining 25% and how will you finish it?"***

> ***Team Answer:*** *"Respected Examiners, our 75% milestone proves the mathematical validity, ensemble architecture, explainability, and interactive UI of our system. For the final 25%, our work is divided into 4 clear engineering milestones: (1) Generating full ROC-AUC curves and sensitivity/specificity matrices across all 5 risk tiers; (2) Integrating acoustic prosody extraction using librosa for complete tri-modal scoring; (3) Containerizing the backend into FastAPI microservices with Docker for hospital EHR integration; and (4) Expanding the questionnaire and crisis lexicon into Hindi and Telugu for rural health accessibility. All 3 members have their modules assigned and we are on track for 100% completion for final defense."*

6. Top 5 Viva Questions & Fast Technical Answers

Q1: Why is this system named 'Smart Health Risk Indicator'?

Ans: It dynamically synthesizes psychometric assessments, physiological sleep habits, lifestyle screen/exercise data, and natural language journal entries into an adaptive 0–100 risk score with built-in uncertainty bounds.

Q2: Why use Monte Carlo Dropout instead of standard predictions?

Ans: Standard neural networks give overconfident point estimates. MC Dropout keeps dropout active at test-time over 20 passes to sample model epistemic uncertainty, yielding a clinical 95% Confidence Interval.

Q3: How does Stacked Generalization differ from simple ensemble averaging?

Ans: Simple averaging assigns static weights. FusionNet is a meta-learner trained on 23 inputs (20 base features + 3 specialist outputs) that dynamically learns non-linear cross-model weighting based on patient context.

Q4: How does your NLP engine handle linguistic negation?

Ans: The NLP parser inspects preceding tokens for negators ('not', 'never', 'can't'). If a positive word like 'happy' is preceded by 'not', it inverts polarity to elevate distress appropriately.

Q5: How does the system protect patient data privacy?

Ans: All neural network scoring runs locally within the client environment. Raw patient reflections and questionnaire responses do not leave the device for inference.